Louis Duvignacq

Machine Learning | Autonomous Systems | Applied Mathematics
Seeking a 6-month end-of-study internship starting in March 2027
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EDUCATION

Master in Drones, Autonomous Systems and AI **2026-2027**
ISAE-SUPAERO, Toulouse, France
- **Core courses**: Perception, Navigation, Planning and Decision for autonomous systems Advanced deep learning, Reinforcement Learning, Big Data and Ethical AI challenges

Master in General Engineering majoring in Applied Mathematics and AI **2023-2027**
École polytechnique, Palaiseau, France
- **Core courses**: Deep Learning, Optimization for AI, Uncertainty quantification & risk analysis, Data Science, Operational Research, Collaborative Learning, Data visualization

***Classe préparatoire* PC* (BSc equivalent) - Lycée Henri-IV, Paris** **2021-2023**

EXPERIENCE

AI Research Intern - Audio Processing and Classification **Mar 2026 - Sep 2026**
Thales Innovations Asia, Singapore
- Developed Transformer and CNN classifiers on urban acoustic scene datasets: **90% accuracy**
- Implemented speech augmentation strategies robust to distribution shift: **+15 pts accuracy**
- Built a clean, reproducible PyTorch training stack (YAML configs, argparse CLI)

Consulting Intern - Industrial Competitiveness **Jun 2025 - Sep 2025**
Avencore, Paris, France
- Supported strategy & performance engagements for industrial leaders in energy and defense
- Built models to assess the competitiveness of several sites and suppliers for top french firms

President of XProjets (Junior Enterprise) **Dec 2024 - Dec 2025**
École polytechnique, Palaiseau, France
- Led the 2024-25 strategic plan: **+30% revenue** (to 220 kEUR) with a team of 20
- Established 3 new corporate partnerships; organized 10 student and corporate events

Gendarmerie Officer - École polytechnique military service **Sep 2023 - Apr 2024**
Compagnie de Gendarmerie, Toulouse-Mirail, France
- Conducted investigation teams (4 gendarmes) on multiple burglary and violent-crime cases

PROJECTS

Scientific project - Probabilistic age estimation with CNNs
- Implemented a PyTorch CNN baseline for apparent age estimation (APPA-REAL): MAE 4 years
- Designed **uncertainty quantification** on a DEX head: **MC-Dropout**, heteroscedastic loss

Research thesis - Statistical analysis of momentum in sports
- Analyzed 1,240 NBA games: momentum tests with **copulas**, **ARMA-GARCH** and **Hawkes processes**
- Applied **Extreme Value Theory** (GEV, GPD) to runs and blowouts; **Monte Carlo** comeback model

Other projects
- **NBA data visualization** - D3.js graphs revealing a team's playstyle from game statistics
- **Loudspeaker design** - analysis of membrane-based speakers through their vibrational modes

SKILLS

- **Programming**: Python (PyTorch, NumPy, pandas, sklearn), SQL, Java, JS (D3.js), HTML/CSS, Git,
- **AI**: Machine learning, deep learning, computer vision, time series, uncertainty quantification
- **Languages**: French (native), English (fluent, TOEFL iBT 109/120), Spanish (B2)

INTERESTS

- **Basketball**: 6 years at regional level, Stade Français first team, certified referee since 2017
- **Piano**: classical & jazz since age 6